

Edrill Bilan

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Education

Pamantasan ng Lungsod ng Maynila BS in Computer Science <i>Tertiary Education - Magna Cum Laude, GWA: 1.35 (Graduation: Aug. 2026)</i>	Sept 2022 – Present
Universidad de Manila STEM Strand <i>Senior High School - Graduated with High Honors, GWA: 95.0</i>	Oct 2020 – July 2022

Skills

Programming Languages: Python, Java, C, SQL, JavaScript, TypeScript, HTML, CSS, x86 Assembly

Machine Learning PyTorch, TensorFlow, ChromaDB, RAG, scikit-learn, Pandas, NumPy, Matplotlib, SciPy

Backend Development: MySQL, PostgreSQL, Django, Flask, ExpressJS, FastAPI, Firebase, Supabase

Frontend & Other Tools: ReactJS, NextJS, Tailwind CSS, JavaFX, Git, Android Studio, Figma, Vercel, Ollama

Experience

National Cheng Kung University <i>Research Intern</i> <i>Computer Vision and Multimedia Laboratory, Tainan, Taiwan</i>	Oct 2025 – Nov 2025
International Research Collaboration with NCKU's Computer Vision and Multimedia Laboratory, finetuning ResNet-50 w/ multi-scale transformers for Pulmonary Lung Nodule Detection on Chest X-ray images as part of undergraduate thesis; published on MDPI's Q1 Journal of Diagnostics.	
Goldilocks Bakeshop, Inc. <i>IT Service Management Intern</i> <i>Main Office, Greenfield Building, Mayflower St. corner Williams, Mandaluyong City</i>	June 2025 – July 2025
Assisted in maintaining IT infrastructure while also coordinating service requests such as technical support for hardware & software issues, streamlining issue resolution workflows across departments.	

Projects

(Machine Learning) OmniJuris – PHL Legal Intelligence RAG System (2026)

- Developed a RAG pipeline over 3.2M chunks of the Philippine OmniCorpus dataset with multilingual e5-base embeddings, ChromaDB vector search, and cross-encoder reranking
- Integrated a locally-hosted Qwen3 4B LLM via Ollama with SQLite-backed conversational memory for context-aware legal queries
- Stack: FastAPI, React, TypeScript, ChromaDB, sentence-transformers, Ollama, SQLite

(Machine Learning) Atlas990 – USA Nonprofit Organizations Intelligence System (2026)

- Cleaned raw public IRS Form 990 tax data into an optimized Parquet format
- Developed a full-stack web application that uses an XGBoost model to calculate a 0–100 priority score for nonprofit leads and integrates Meta's FAISS library to instantly find matching lookalike organizations.
- Stack: React, Vite, TypeScript, FastAPI, Pandas, NumPy, scikit-learn, XGBoost, FAISS

(Deep Learning) Pulmonary Lung Nodule Detection System (2026)

- Developed, trained, and finetuned a ResNet-50 deep learning model enhanced with light transformer blocks per stage for early lung cancer screening on chest X-rays as part of undergraduate thesis.
- Stack: Python, PyTorch, OpenCV, Pandas, NumPy

(Web Crawler) TrabaHound (2026)

- Developed a comprehensive job aggregation platform featuring a Next.js and TypeScript frontend, serverless APIs, and a Python crawler Jobspy.
- Implemented PDF parsing and LLM-based resume analysis for enhanced job matching.
- Stack: NextJS, TypeScript, Python, Jobspy, LLMs, PDF Parsing

(Full-Stack) MAFA-Inventi (2025)

- Co-developer for a property management system developed for the Inventi Hackathon enabling property managers to upload floor plans, pin issues, and efficiently manage properties, tenants, and maintenance.
- Implemented an LLM-based dashboard analysis.
- Stack: ReactJS, Tailwind CSS, NodeJS, ExpressJS, Supabase, Vercel

(Full-Stack) Fantasy Flip Memory Game Android Application (2024)

- Lead developer for an Android mobile game, handling both frontend and backend logic.
- Stack: Java, XML, Firebase Realtime Database, Android Studio

(Full-Stack) DOM Compiler (2024)

- Lead developer for the DOM programming language, designed with custom personalized syntax and keywords to meet Compiler Design course requirements.
- Stack: NextJS, Tailwind CSS, Flask, React, NodeJS

Extracurricular

- **Data Engineering Pilipinas | DataCamp Scholar**

DataCamp x Data Engineering Pilipinas Partnership Scholar 2026

- **Inventi Asia Hackathon | Creativity Award Winner**

Smart Property Solutions Hackathon 2025

- **Google Developer Student Club (GDSC) Volunteer | Web Development Team**

October 2024 – October 2025

- **Google Developer Student Club (GDSC) Volunteer | Design and Branding Team**

November 2023 – October 2024

- **DSPC Annual Journalism Competition | 3rd Place Winner in Online Publishing (English)**

Manila Division Schools Press Conference 2019

Publications

Bilan, E. F., Manduriaga, E. T., et al. (2026). RNNet-MST: A ResNet-50 with Multi-Scale Transformer Blocks for Pulmonary Nodule Classification and Attention-Based Localization on Chest X-Ray Images. *Diagnostics*, 16(10), 1574. DOI: 10.3390/diagnostics16101574

- Undergraduate thesis developed through a PLM–NCKU international research collaboration with the Computer Vision and Multimedia Laboratory at National Cheng Kung University, Taiwan.
- Published in MDPI Diagnostics, a Q1 peer-reviewed journal, indexed in Web of Science and Scopus.

References

Dr. Hernando Salapare III, *Researcher at CNRS - IS2M, France*

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Bren Alfred España, *Business Solutions Manager, JTI*

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